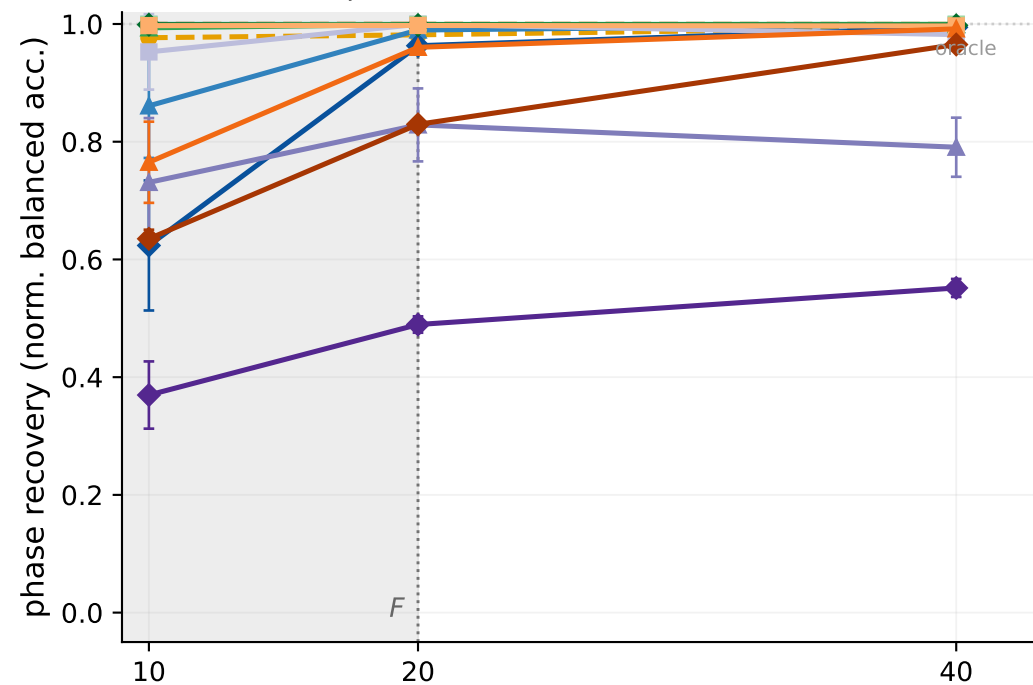
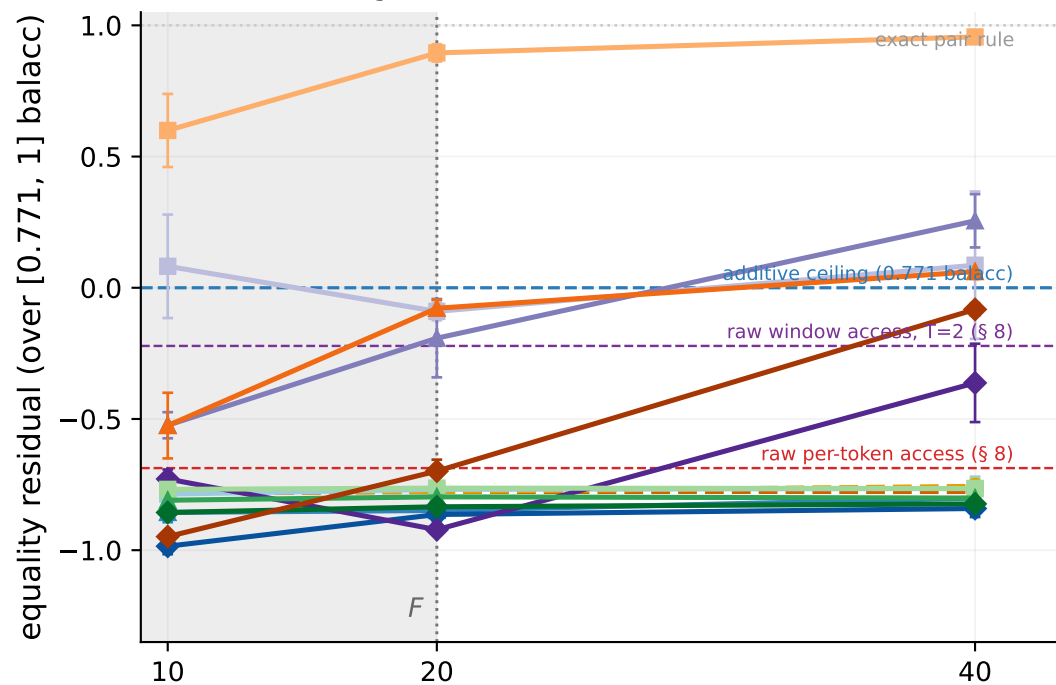


(a) DC control: phase class c_t (b) PRIMARY: regime-3 residual of $e_t = [c_t = c_{t-1}]$ dictionary size d_{sae} (per-token)

T-SAE (per-token)

TXC-pre (T=2)

TXC-pre (T=4)

TXC-pre (T=8)

TXC-post (T=2)

TXC-post (T=4)

TXC-post (T=8)

Stacked-SAE (T=2)

Stacked-SAE (T=4)

Stacked-SAE (T=8)

Spectral-TXC (T=2)

Spectral-TXC (T=4)

Spectral-TXC (T=8)